From Prediction to Persuasion: Agentic Recommendation Reason Generation for Regulatory-Compliant Financial AI

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Abstract. Financial product recommendation systems must explain *why* a product is recommended — first and foremost because persuasion requires narrative, not probability: a customer who asks "Why should I buy this fund?" needs a reason they can accept, not a score they must trust. Regulators (Korean FSS, EU AI Act) are one important audience for such explanations, but the primary driver is the human act of persuasion itself. We present a three-stage pipeline that bridges the gap between model prediction and human persuasion: (1) IG-guided knowledge distillation from a heterogeneous-expert PLE teacher to per-task LGBM students, preserving explanation-relevant features while enabling GPU-free CPU inference; (2) a multi-agent recommendation reason generation pipeline where three specialized agents (Feature Selector, Reason Generator, Safety Gate) collaboratively produce natural-language explanations grounded in business-mapped feature attributions; (3) regulatory compliance by design, with built-in drift monitoring, fairness auditing, and governance reporting aligned to Korean FSS guidelines, the EU AI Act, and the Korean AI Basic Act. We evaluate distillation quality (AUC gap < X%), reason generation quality via human evaluation (N domain experts, Y% preference over template baselines), and Safety Gate reliability (precision/recall). The system achieves sub-100ms serving latency on AWS Lambda at a fraction of the cost of dedicated GPU inference servers.

Keywords: Recommendation explanation, Knowledge distillation, LLM agents, Regulatory compliance, Financial AI, EU AI Act

1 Introduction

1.1 The Persuasion Problem

The final deliverable of a financial recommendation system is not a probability but a *reason the customer can accept*. A model outputting $P(\text{invest} \mid x) = 0.73$ provides no value to:

- The **customer** who asks "Why should I buy this fund?"
- The **relationship manager** who needs a talking point for the sales call.
- The **regulator** who demands "Why was this decision made?" under Article 13 of the EU AI Act.

Existing explanation approaches are insufficient:

- **SHAP/LIME**: "feature_237 contributed 0.12" — no business meaning, unstable under perturbation [7, 10].
- **Template-based**: "Customers like you also bought X" — rigid, ignores individual context.
- **Direct LLM generation**: Hallucination risk, no grounding in actual model reasoning, regulatory non-compliance.

1.2 Our Approach

We propose a full-chain solution from prediction to persuasion:

1. **Knowledge Distillation**: A heterogeneous-expert PLE teacher (18 tasks, 7 experts, 318 features; organized by the companion paper's reductionist two-axis framework of Financial DNA × Data Modality) is distilled into per-task LGBM students using IG-guided feature selection. This enables GPU-free serving while preserving the features that matter for explanation.
2. **Multi-Agent Reason Generation**: Three specialized LLM agents collaborate in a pipeline — Feature Selector chooses explanation-worthy features, Reason Generator produces natural-language narratives, and Safety Gate validates regulatory compliance.
3. **Compliance by Design**: Drift monitoring, fairness auditing, audit trails, and governance reporting are embedded in the pipeline, not bolted on after deployment.

1.3 Contributions

1. **IG-guided Distillation with Explanation Preservation**: Feature selection optimizes not only

for prediction accuracy but also for explanation material availability.

2. **Feature Business Reverse-Mapping**: A systematic registry (`interpretation_registry`) that maps every feature to a business-interpretable description, enabling grounded explanation generation.
3. **3-Agent Reason Generation Pipeline**: Role-separated agents (selection → generation → safety) with independent improvement and audit logging.
4. **Safety Gate for Financial Compliance**: Automated checking for hallucination, inappropriate investment advice, and regulatory violations (Financial Consumer Protection Act (금소법), Suitability Principle (적합성 원칙)).
5. **Regulatory Compliance Architecture**: Explicit mapping of system components to Korean FSS guidelines, EU AI Act articles, and the Korean AI Basic Act.
6. **Human Evaluation Protocol**: Systematic evaluation of recommendation reason quality by domain experts.

2 Related Work

2.1 Knowledge Distillation for Recommendation

Hinton et al. [5] established the teacher-student paradigm via temperature-scaled soft labels. In recommendation, distillation has been applied to compress deep models into lightweight servers. CKD (Collaborative Knowledge Distillation) and ranking distillation preserve recommendation-specific structure.

Gap: No prior work selects distillation features based on *explanation value* alongside predictive value.

2.2 Recommendation Explanation Generation

Recommendation explanation approaches fall into three categories:

Template-based methods map model outputs to prewritten phrases ("Recommended because you recently viewed similar products"). Amazon and Netflix use variants of this approach at scale. Templates are safe and fast but lack personalization and flexibility — the same template serves millions of different customer contexts.

Attention-based methods use model-internal attention weights as feature attributions for explanation. While more dynamic than templates, attention weights are contested as reliable explanations, and the resulting attributions still require translation into business language.

LLM-based generation is an emerging approach that uses large language models to produce natural-language explanations from model outputs. Recent work explores LLM-augmented recommendation explanation, but faces challenges of hallucination, regulatory compliance, and grounding in actual model reasoning.

Gap: No system combines structural model explainability (gate weights) with multi-agent LLM-based generation under regulatory safety constraints, where each agent has a specialized role (selection, generation, validation) and all outputs are audit-logged.

2.3 Responsible AI in Finance

Korean FSS published AI guidelines (2021) [3] and model risk management directives requiring explainability, fairness monitoring, and audit trails. The EU AI Act [2] classifies financial credit/recommendation as high-risk AI, mandating transparency (Art. 13), human oversight (Art. 14), and accuracy (Art. 15). The EBA [1] calls for "interpretable" models in internal risk assessments. Korea's AI Basic Act [8] (December 2024) adds domestic high-risk AI classification.

Pearl [9] argues that true explanation requires causal understanding, not mere statistical association — a position increasingly echoed by financial regulators who demand explanations reflecting the actual decision mechanism [11].

Gap: No recommendation system provides an explicit, verifiable mapping from regulatory requirements to system architecture components, with explanations grounded in causal reasoning rather than post-hoc correlation.

2.4 LLM Safety and Grounding

Deploying LLMs for customer-facing financial text introduces specific risks: hallucination (stating non-existent product features), inappropriate advice (recommending unsuitable products for the customer's risk profile), and regulatory violation (breaching 금소법 or 적합성 원칙).

Retrieval-Augmented Generation (RAG) mitigates hallucination by grounding generation in retrieved factual context. Our approach extends this: rather than retrieving from a general knowledge base, we ground generation in *model-internal feature attributions* that have been reverse-mapped to business descriptions. This ensures the generated text reflects what the model actually computed, not what an LLM independently "knows."

Content filtering and safety gates provide a final defense layer. Our 3-agent architecture separates generation from validation, enabling independent improvement of each component.

3 Knowledge Distillation

3.1 Teacher-Student Architecture

Distillation diagram placeholder
PLE Teacher (GPU, weekly) → Soft Labels → LGBM Students (CPU, daily)
1. IG feature selection

Figure 1: Teacher-student distillation architecture.

The teacher model (PLE with 7 heterogeneous experts, 18 tasks, 318 features; see companion paper for architecture details) produces soft probability outputs that serve as training targets for per-task LGBM [6] students. The teacher's value as a distillation source stems from its *structural guarantee* against *expert collapse*: because the seven experts are architecturally distinct (DeepFM, Mamba, HGCN, PersLay, etc.), they cannot converge to the same function, ensuring the soft labels encode genuinely multi-faceted customer understanding.

The key design decision is *per-task distillation*: rather than a single student model for all 18 tasks, we train 18 independent LGBM models, each learning one task's soft labels. This enables: (1) per-task feature selection (different tasks benefit from different features), (2) independent retraining (if one task drifts, only its student is re-distilled), (3) interpretable feature importance per task (LGBM's built-in feature importance aligns with the business reverse-mapping for explanation generation).

Lifecycle separation:

- **Teacher**: retrained weekly/monthly on SageMaker (GPU required, comprehensive). The teacher captures complex inter-task relationships via adaTT and non-linear expert interactions that LGBM cannot directly learn.
- **Students**: re-distilled daily with fresh soft labels (CPU only, fast). Daily re-distillation tracks data drift without the cost of GPU training.
- **Champion-Challenger**: automatic comparison of new student vs. current production model. If the new student's AUC drops below threshold, the update is blocked and an alert is raised.

This architecture resolves a fundamental tension in financial AI: the *model that learns best* (deep PLE with GPU) is not the *model that serves best* (lightweight LGBM on CPU Lambda). Knowledge distillation [5] bridges this gap, and the FD-TVS (Friedman Decomposition–Transitory Variance Scoring) system [4] further weights the student's predictions by income stability type. FD-TVS applies Friedman's Permanent Income Hypothesis to decompose each customer's observed income into permanent (stable, long-term) and transitory (bonus, irregular) components via HP or Kalman filtering. Customers whose income is predominantly transitory receive lower confidence weights for long-horizon product recommendations, preventing the system from recommending products that assume stable cash flow to customers with volatile income patterns.

3.2 IG-based Feature Selection

Integrated Gradients [12] computes per-feature attribution from the teacher model. We select top-k features ranked by IG importance, with a dual objective:

$$\text{score}(f) = \alpha \cdot \text{IG}_{\text{pred}(f)} + (1 - \alpha) \cdot \text{IG}_{\text{explain}(f)}$$

where IG_{pred} measures predictive contribution and $\text{IG}_{\text{explain}}$ measures the feature's value as explanation material.

The explanation score $\text{IG}_{\text{explain}(f)}$ is derived from the feature's reverse-mapping richness in the interpretation registry: features with detailed business descriptions, clear directionality, and natural-language templates receive higher scores. For example, `hmm_lifecycle_prob_growing` (explanation-rich: "customer is in growth stage") scores higher than `mamba_temporal_d17` (explanation-poor: generic embedding dimension).

This dual-objective selection ensures that the student model retains not only the most predictive features but also the features that generate the most compelling recommendation reasons. The hyperparameter α controls the trade-off: $\alpha = 1$ optimizes purely for prediction, $\alpha = 0$ purely for explanation quality. We empirically find $\alpha = 0.7$ balances both objectives.

The resulting feature set is typically 40–80 features per task (down from 318), achieving >95% of teacher AUC while providing sufficient explanation vocabulary. Critically, the dual-objective selection preserves features across all four Financial DNA dimensions (engagement, lifecycle, value, consumption) and multiple Data Modalities (temporal, graph, topological, etc.) from the com-

panion paper's two-axis decomposition, ensuring the student retains multi-faceted understanding rather than collapsing to a single dominant feature group.

3.3 Distillation Results

Task	Teacher AUC	Student AUC	Gap
has_nba	–	–	–
churn_signal	–	–	–
will_acquire_deposits	–	–	–
will_acquire_investments	–	–	–
...	...	...	...

Table 1: Distillation results per task (TODO).

The `interpretation_registry` serves as the **data contract** between distillation and reason generation. Distillation (Section 3) decides *which* features to preserve in the student model — optimizing for both predictive accuracy and explanation material via the dual-objective IG score. Reason generation (Section 4) decides *how* to explain those features to humans — mapping each preserved feature to business-interpretable language. This separation means the two stages can evolve independently: improving feature selection does not require rewriting explanation templates, and vice versa.

4 Recommendation Reason Generation

4.1 Feature Business Reverse-Mapping

Every feature in the system is registered in an `interpretation_registry` with structured business metadata:

Feature	Business Mapping	Explanation Template
hmm_lifecycle_prob_growth	Growth stage probability	"Your asset portfolio is in a growth phase"
mamba_temporal_13	13-month spending trend	"Your spending has been increasing recently"
hgcn_hierarchy_dist	Product category position	"Investment products are a natural next step"
synth_stability	Transaction stability	"You maintain a stable transaction pattern"
gmm_cluster_prob_3	Segment probability	"You share characteristics with active investors"

Table 2: Feature reverse-mapping examples.

This registry serves dual purposes: (1) grounding material for the Reason Generator agent, and (2) audit trail showing which features influenced each recommendation.

4.2 3-Agent Pipeline Architecture

3-Agent pipeline diagram placeholder

Agent 1: Feature Selector
(IG attribution → top features → explanation-worthy supplementation)
↓
Agent 2: Reason Generator
(reverse-mapped contexts → natural language reason)
↓
Agent 3: Safety Gate
(regulatory check → hallucination check → appropriateness check)
↓
Pass → Serve | Fail → Template fallback

Figure 2: Three-agent recommendation reason generation pipeline.

4.2.1 Agent 1: Feature Selector

Selects features for explanation based on:

- IG attribution scores (model-driven relevance)
- Business reverse-mapping richness (explanation-driven relevance)

- Customer context (personalization: different features matter for different customer profiles)

The Feature Selector uses the Financial DNA axis from the companion paper's two-axis framework to select features from the relevant customer dimension (e.g., lifecycle features for a churn recommendation, value features for an investment recommendation), ensuring explanations address the dimension most pertinent to the recommended action.

4.2.2 Agent 2: Reason Generator

Receives selected features with their business reverse-mappings and generates a natural-language recommendation reason. The Reason Generator structures its narrative around Financial DNA groups — opening with the customer's current state (lifecycle), grounding the recommendation in observed behavior (engagement/consumption), and framing the value proposition in terms the customer recognizes (value dimension) — producing what the companion paper calls *multi-faceted customer understanding* rendered as persuasive text. Grounding constraints:

- Must reference only features actually present in the selection.
- Must use business terminology appropriate for the target audience (customer vs. relationship manager).
- Must follow financial domain tone and compliance guidelines.

Example output (customer-facing):
"고객님은 현재 자산 성장 단계에 있으시고, 최근 3개월간 소비가 증가하는 추세입니다. 보유하신 상품 구조상 투자 펀드가 자연스러운 다음 단계이며, 안정적인 거래 패턴을 고려할 때 중위험 균형형 펀드를 추천드립니다."

Figure 3: Generated recommendation reason example (Korean, customer-facing).

4.2.3 Agent 3: Safety Gate

Validates the generated reason against:

Check Category	Criteria
Hallucination	No claims about facts not in feature data
Regulatory	No violation of 금소법, 적합성 원칙
Appropriateness	No unsuitable investment advice for risk profile
Tone	Professional financial advisory language
Factual	Numerical claims match actual feature values

Table 3: Safety Gate validation criteria.

On failure: automatic fallback to template-based safe reason. All gate decisions are logged for audit trail.

4.3 Caching Strategy

Recommendation reasons are cached by customer segment × product category × feature pattern hash. Cache hit avoids LLM invocation entirely, reducing latency and cost.

5 Regulatory Compliance

5.1 Korean FSS Guidelines Mapping

FSS Requirement	System Component	Verification
Explainability	Gate weights + 3-agent reason	Per-recommendation audit log
Fairness	FairnessMonitor (DI/SPD/EOD)	Weekly automated report
Model validation	Champion-Challenger	Pre-deployment comparison
Monitoring	DriftDetector (PSI)	Continuous, 3-day trigger
Audit trail	HMAC hash-chain logs	Immutable, 7 audit tables
Fallback	Template reason + kill switch	Instant manual override

Table 4: Korean FSS guideline compliance mapping.

5.2 EU AI Act Mapping

Article	Requirement	System Component
Art. 13	Transparency	3-agent reason generation + feature attribution
Art. 14	Human oversight	Human-in-the-Loop review + kill switch
Art. 15	Accuracy & robustness	Ablation-validated degradation + drift monitoring
GDPR Art. 22	Right to opt-out	Consent/opt-out audit table + manual override

Table 5: EU AI Act article-level compliance mapping.

5.3 Korean AI Basic Act (2024.12)

5.4 Monitoring and Governance

Monitoring architecture placeholder

DriftDetector (PSI) → FairnessMonitor (DI/SPD/EOD) → HerdingDetector (HHI/Gini)
↓
IncidentReporter → GovernanceReportGenerator
↓
Monthly/quarterly automated report

Figure 4: Monitoring and governance architecture.

5.4.1 Human-in-the-Loop

Regulatory bodies (Korean FSS, EU AI Act Art. 14) require human oversight. The system implements this at multiple levels:

- **Reason sampling review**: Periodic human review of generated reasons.
- **Model replacement approval**: Champion-Challenger results require human sign-off.
- **Incident escalation**: Automated anomaly detection triggers human investigation.
- **Fairness review**: Periodic human audit of fairness metrics.

6 Experiments

6.1 Distillation Experiments

6.2 Reason Generation Quality

6.2.1 Human Evaluation Protocol

- **Evaluators**: N financial domain experts (bank product managers, compliance officers).
- **Evaluation criteria**: Accuracy (1-5), Naturalness (1-5), Persuasiveness (1-5), Regulatory fitness (1-5).
- **Comparison**: (A) Template-based, (B) SHAP-based + template, (C) 3-agent pipeline.
- **Method**: Blind evaluation, each evaluator rates 100 recommendation reasons.

Method	Accuracy	Natural.	Persuasion	Compliance
Template	–	–	–	–
SHAP+Template	–	–	–	–
3-Agent (ours)	–	–	–	–

Table 6: Human evaluation results (TODO).

6.3 Safety Gate Evaluation

Check	Precision	Recall	Fallback Rate
Hallucination	–	–	–
Regulatory	–	–	–
Appropriateness	–	–	–
Overall	–	–	–

Table 7: Safety Gate precision and recall (TODO).

6.4 Serving Performance

Component	Latency	Cost/1K req	Notes
LGBM inference	< 10ms	–	Lambda CPU
IG attribution	< 50ms	–	Lambda CPU
Reason generation	< 200ms	–	Cache hit: < 5ms
Safety Gate	< 50ms	–	Rule-based + LLM
Total	< 300ms	–	Cache hit: < 100ms

Table 8: Serving latency breakdown (Lambda serverless, no GPU).

6.5 Regulatory Compliance Audit

7 Discussion

7.1 Findings Summary

7.2 The Dual Role of Features

A key insight from this work: features serve dual purposes in financial recommendation. Even features with marginal predictive contribution (e.g., TDA topological features may add only $\Delta\text{AUC} = 0.01$) provide irreplaceable context for recommendation reasoning. Internally, TDA persistence captures behavioral shape stability — but the customer never sees "persistent homology" or "Betti numbers." Instead, the `interpretation_registry` reverse-maps this to business language (e.g., "꾸준한 거래 패턴을 유지하고 계십니다"), and the LLM agent weaves it into a natural-language reason.

This reframes feature engineering evaluation: the value of a feature is not solely its predictive contribution but also its contribution to the recommendation context available to the reason generation pipeline. As argued in the companion paper, *what to observe* matters more than *how to model* — features derived from domain-specific questions ("Is their income permanent or transitory?", "Is product adoption spreading like contagion?") enrich the internal reasoning context, enabling more nuanced business-language explanations than any amount of architectural sophistication applied to shallow statistical summaries.

7.3 Practical Deployment Considerations

- **LLM selection**: On-premises LLM vs API (latency, cost, data residency).
- **Reason quality maintenance**: Periodic human review + automated quality scoring.
- **Regulatory updates**: Architecture supports adding new compliance checks without redesign.

7.4 Limitations

- LLM dependency introduces cost, latency, and residual hallucination risk.
- Human evaluation scale limited by expert availability.
- Korean/EU regulatory focus; other jurisdictions not yet analyzed.
- Template fallback quality is inherently limited.

7.5 Future Work

- Real customer A/B testing of reason quality → conversion rate.
- Multi-lingual reason generation (Korean, English, Chinese).
- Automated regulatory update pipeline (regulation change → compliance check update).
- Fine-tuned domain-specific small LLM to replace general-purpose API.

8 Conclusion

We presented a full-chain system that bridges the gap between model prediction and human persuasion in financial product recommendation.

Three key contributions define this work. First, IG-guided knowledge distillation with a dual-objective feature selection preserves both predictive accuracy and explanation material when compressing a complex PLE teacher into lightweight LGBM students. Second, the 3-agent recommendation reason generation pipeline (Feature Selector → Reason Generator → Safety Gate) produces natural-language explanations grounded in business-mapped feature attributions, with role separation enabling independent improvement and audit logging. Third, regulatory compliance is embedded by design — Korean FSS guidelines, the EU AI Act, and the Korean AI Basic Act are explicitly mapped to system architecture components, with automated monitoring (drift, fairness, herding) and human-in-the-loop oversight at critical decision points.

The fundamental insight is that features serve a dual role in financial AI: they contribute to prediction *and* to the explanation vocabulary that ultimately persuades customers, empowers relationship managers, and satisfies regulators. This reframes the traditional feature engineering calculus: a feature with marginal AUC contribution but rich business interpretability may be more valuable than a high-AUC feature that generates no meaningful explanation.

The system is designed for deployment on serverless infrastructure (AWS Lambda), achieving sub-100ms serving latency without dedicated GPU servers — matching the operational reality of financial institutions with limited ML engineering resources.

Author Contributions

Author 1 (PM/Data Scientist): Designed the distillation strategy, recommendation reason generation pipeline, and regulatory compliance architecture. Led the overall system design based on domain expertise in financial risk management. Authors 2 and 3: Implemented the distillation pipeline, serving infrastructure, and monitoring components as development supporters.

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